MIL-P-2911A
17 April 1957
SUPERSEDING
MIL-P-2911
4 September 1951

MILITARY SPECIFICATION

PACKING ASSEMBLY, HYDRAULIC CONICAL AND V TYPES

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers packing assemblies for hydraulic fluid service such as elevators, hoists, and steering gears.

1.2 Classification. Hydraulic packing assemblies shall be of the following types, as specified (see 6.1):

Type I — Hydraulic conical, symbol 1210.

Type II — Hydraulic V, symbol 1220.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

LLL-B-631 — Boxes, Fiber, Corrugated (for Domestic Shipment).

LLL-B-636 — Boxes, Fiber, Solid (for Domestic Shipment).

PPP-B-585 — Boxes, Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

MILITARY

JAN-P-108 — Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

MIL-P-116 — Preservation, Methods of.

MIL-B-121 — Barrier Material, Greaseproof, Flexible (Waterproofed).

MIL-B-10377 — Boxes, Wood-Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Liners, Case, Waterproof.

FSC 5330

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:53Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Qualification. Packing material furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.2).

3.2 Composition and construction. The composition and construction shall conform to that of the sample given qualification approval (see 4.1).

3.3 Size. Packing shall be furnished in the sizes specified (see 6.1).

3.4 Graphite lubrication. Graphite lubrication shall be applied to the surface of each packing ring, and shall not exceed 2 percent, by weight, of the finished packing.

3.5 Friction compound. The friction compound shall be not more than 60 percent, by weight, of the finished packing. The compound shall contain not less than 20 percent by weight, of synthetic rubber.

3.6 Simulated performance. The packing, during initial conditions and during progress of the simulated performance test specified in 4.4.2, shall respond readily to gland adjustment for the control of leakage from the stuffing box. The packing shall be in satisfactory condition upon completion of test.

3.7 Type I, hydraulic conical.

3.7.1 *Construction.* Packing shall consist of conical or beveled shape rings which permit each ring to nest within the adjacent ring, and one top and bottom adapter ring. The ring and adapters shall be composed of layers of woven cotton fabric, and treated with a synthetic rubber friction compound.

3.7.2 *Fabric.* The cotton fabric shall be not less than 40 percent by weight.

3.8 Type II, hydraulic V.

3.8.1 *Construction.* Packing shall consist of either V, U, or C cross-sectional shape rings which permit each ring to nest within the adjacent ring, and one top and bottom adapter ring. The rings and adapters shall be composed of layers of either cotton, asbestos or wire-inserted asbestos fabric, and treated with a synthetic rubber friction compound. Where wire-inserted asbestos cloth is used in the construction, it shall be so placed that it will bear against the wall of the stuffing box and not against the moving member.

3.8.2 *Fabric.* The cotton, asbestos, and wire-inserted asbestos fabric shall be not less than 40 percent, by weight.

3.9 Workmanship. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests at a Government laboratory. Qualification tests shall be conducted

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:53Z
Check the source to verify that this is the current version before use.

at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.4.

4.2 Sampling.

4.2.1 *Lot.* For purposes of sampling, a lot shall consist of all packing of one type and size produced in one plant under essentially the same conditions and offered for delivery at one time.

4.2.2 *Sampling for visual and dimensional inspection.* A random sample of sets shall be selected from each lot of material offered for Government inspection in accordance with Standard MIL–STD–105 at inspection level III, and shall be subjected to the inspection specified in 4.3.1. The acceptable quality level shall be 2.5 percent defective.

4.2.3 *Sampling for lot acceptance tests.* A random sample of sets shall be selected from each lot offered for Government tests in accordance with the appendix to Standard MIL–STD–105 at inspection level L–4. In addition a 12-inch square sample of untreated cotton or asbestos fabric shall be selected from each lot used in the manufacture of the finished packing. The sample sets and 12-inch square shall be forwarded to the Government laboratory for testing specified in 4.3.2.

4.3 Inspection and tests.

4.3.1 *Visual and dimensional inspection.* Each of the sample sets selected in accordance with 4.2.2 shall be visually and dimensionally inspected to verify compliance with the requirements of this specification. Any set in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective sets in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.2 *Lot acceptance tests at Government laboratory.* The samples selected in accordance with 4.2.3 shall be subjected to those tests deemed necessary by the Government laboratory to determine that the samples conform with that given qualification. If any sample set, or the twelve-inch square, fails any test the lot represented by the sample shall be rejected.

4.3.2.1 *Place of tests.* Unless otherwise specified in the contract or order, tests shall be conducted at the U. S. Naval Engineering Experiment Station, Annapolis, Md.

4.3.3 *Rejected lots.* Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming material. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

4.4 Test procedures.

4.4.1 *Chemical analysis.* Chemical analysis shall be made using standardized procedures.

4.4.2 *Simulated performance (see 3.6).* Performance tests shall be made under simulated operating conditions in a machine designed for the purpose, as follows:

4.4.2.1 The apparatus consists of a $6\frac{1}{2}$-inch diameter hydraulic ram fitted with a cylinder at one end and a $\frac{5}{8}$-inch diameter valve stem at the other end. The ram has a stroke of 6 inches and makes 12 complete forward and return cycles per minute. The valve stem has a stroke of 2 inches and makes 12 complete forward and return cycles per minute. The packing shall be initially tested for tightness under hydrostatic oil pressure of 3,500 pounds per square inch gage, the rams and valves at rest. The packings are then subjected to lubricating oil (symbol 2110) at 210°F., and a pressure of approximately 1,650 pounds per square inch gage. The change from maximum to minimum pressures occurs every stroke of the ram. The apparatus is operated for a total of 1,000 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:53Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing requirements specified herein shall be used to prepare the equipment for level A, B, or C, as specified (see 6.1).

5.1.1 *Preservation and packaging (see 6.1).*

5.1.1.1 *Level A.* The packing shall be packaged in accordance with method 1C-1 or 1C-2 of Specification MIL-P-116. The packing shall be wrapped in greaseproof paper conforming to grade A of MIL-B-121.

5.1.1.2 *Level C.* The packing shall be packaged in accordance with the manufacturer's commercial practice.

5.1.2 *Packing.*

5.1.2.1 *Level A.* Unless otherwise specified in the contract or order, packing shall be packed in wood cleated fiberboard, nailed wood, wirebound, fiberboard, wood cleated paper overlaid (overseas type), or wood cleated plywood (overseas type) boxes conforming to Specifications PPP-B-591 (overseas type), PPP-B-621 (class 2), PPP-B-585 (class 2 or 3), JAN-P-108, MIL-B-10377, and PPP-B-601, respectively. Shipping containers shall have case liners conforming to Specification MIL-L-10547 and appendix thereto. Case liners for boxes conforming to Specification JAN-P-108 may be omitted provided all joints of the boxes are sealed with tape as specified in the appendix of the box specification. Box closures shall be as specified in the appendix of the applicable box specification. The gross weight of wood boxes shall not exceed 200 pounds; that of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.1.2.2 *Level B.* Unless otherwise specified in the contract or order, packing shall be packed in wood cleated fiberboard, nailed wood, wirebound, corrugated or solid fiberboard, wood cleated plywood (domestic type), or wood cleated paper overlaid (domestic type) boxes conforming to Specifications PPP-B-591 (domestic type), PPP-B-621 (class 1), PPP-B-585 (class 1), LLL-B-631, LLL-B-636, PPP-B-601, and MIL-B-10377, respectively. Closures shall be as specified in the applicable box specification, or appendix thereto. Fiberboard boxes shall conform to the special requirements of the applicable box specification. The gross weight of wood boxes shall not exceed 200 pounds, that of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.1.2.3 *Level C.* Packing shall be packed in containers in a manner to insure safe delivery and acceptance at destination. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.2 **Marking.** In addition to any special marking specified in the contract or order, shipments shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type and size required (see 1.2 and 3.3).

(c) Whether packing shall be packaged in accordance with level A or C, and packed in accordance with level A, B, or C (see 5.1).

6.2 **Qualification.** With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-2911, whether or not such products have actually been so listed by that date.

6.2.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that

Source: https://assist.dla.mil -- Downloaded: 20261O-16T15:53Z
Check the source to verify that this is the current version before use.

they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C.

6.3 The packaging, packing, and marking specified herein apply only to direct shipments to the Government and are not intended to apply to contracts or orders between the manufacturer and the prime contractor.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Ordnance Corps
Navy—Bureau of Ships

Other interest:
Army—ET
Navy—Or

Preparing activity:
Navy—Bureau of Ships

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:53Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

FFICIAL BUSINESS
:NALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Ship Engineering Center
Center Building-Sec 61247
Prince George's Center
Hyattsville, MD 20782

Source: https://assist.dla.mil -- Downloaded: 202610-16T15:53Z
Check the source to verify that this is the currentversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 202610-16T15:53Z
Check the source to verify that this is the current version before use.